Grids and Simulation

Topic 10 of 10

6 questions | skeleton and writing space only

Each question: the prompt, the function or class signature, and ruled lines to write your own solution.

1. Tic-Tac-Toe System with K Players and 3-in-a-Row Win

MEDIUM | asked 2026-05-18 | object-design, grid, simulation

Design a Tic-Tac-Toe system for k players on an n x n board. After each move, return the game status; any player wins immediately after forming 3 consecutive marks in a straight line. Input parameters: k players and an n x n board. Players take turns placing their marks on empty cells. A player wins as soon as they form 3 consecutive marks in a straight line, regardless of how large n is. After each move, return the correct game status, such as Game Over or <player> won.

```
class TicTacToe:

    def __init__(self, n, k, target=3):

    def inside(self, r, c):

    def run_length(self, r, c, dr, dc):

    def move(self, player, r, c):
```

2. Gate Entry / Exit Time Resolution with Tie-Break Rules

MEDIUM | asked 2026-05-04 | simulation, algorithm

Onsite coding round 2: given an array indexed by person id, each containing (timestamp, action in {enter, exit}), return per-person the actual time they passed through the gate after applying tie-break rules between simultaneous enters and exits.

```
def gate_pass_times(events, idle_first='exit'):
```

3. Babylon Tile-Stack Merge Game

HARD | asked 2025-12-29 | simulation, data-structure

Onsite NG round 1 (Munich office): implement a two-player tile-stack merge game ("Babylon") given the rules. Start state is 12 height-1 stacks of 4 colors (Y / W / G / B), 3 of each; a move places one stack on top of another under a constraint; the player who cannot move loses. 12 tiles initially, partitioned into 12 height-1 stacks. 4 colors: Y / W / G / B; exactly 3 tiles per color. Two players alternate moves.

```
def wins(stacks):
```

4. Count Overlapping Rectangle Updates on a Grid

HARD | asked 2026-07-10 |

 # Count Overlapping Rectangle Updates on a Grid Implement range_add_counts(n, rectangles). Start with an n x n matrix of zeroes. Each rectangle is represented as [row1, col1, row2, col2] with inclusive, zero-based coordinates. For every rectangle, add one to each matrix cell inside that rectangle. Return the final matrix, where each cell therefore contains the number of input rectangles covering it.

```
def range_add_counts(n, rects):
```

5. Simulate In-Place Cellular State Updates

HARD | asked 2026-05-04 |

Overview: This question evaluates a candidate's ability to implement grid-based state transitions and manage in-place updates with strict space constraints, assessing skills in matrix manipulation, neighbor traversal, and correctness under simultaneous update rules.

```
def step_in_place(board, wrap=False):
```

6. Count same-color squares in a character grid

HARD | asked 2025-11-22 |

You are given a 2D grid (matrix) of characters. Each character represents a color: cells with the same character are considered the same color. Formally: - The grid has m rows and n columns. - grid[i][j] is an alphabetic character (e.g., 'a'-'z' or 'A'-'Z'). A **square in this grid is defined as a contiguous k x k submatrix aligned with the grid axes, for some integer k >= 1.**

```
def count_same_color_squares(grid):
```